

characteristics are shown in Table1; 11/35 patients received continuous hemofiltration. At T0, there were no differences in SOFA score, hemodynamic variables, NE dose, PaO₂/FiO₂, ScvO₂, time from diagnosis or furosemide dose between patients with and without HT (Table 1). At T24, the degree of NFB did not differ significantly between the group with and without HT (1.7 (1.2-2.0) vs 1.3 (1.1-1.9) L_p=0.1). Patients who tolerated a NFB had lower lactate levels (1.1 (0.7-1.2) vs 1.4 (1.0-1.7) mmol/L, $p=0.03$) and higher SBF (151 (81-194) vs 70 (45-135) PU, $p=0.01$) than those who did not. MAP, cardiac index (CI) and ScvO₂ were similar in patients with and without HT. SBF decreased in patients with and without HT ($p<0.01$) but was higher at H24 in patients who tolerated the NFB than in those who did not (100 (35-173) vs 35 (22-78) PU, $p=0.01$). The area under the ROC curve to predict HT after a NFB was higher for baseline SBF (0.92[0.83-0.99]) than for lactate level (0.72 [0.58-0.87]) and ScvO₂ (0.69 [0.53-0.85]) (both $p=0.03$), with a cut-off ≥ 125 PU (sensitivity 90, specificity 93%).

CONCLUSION. Baseline SBF can predict good hemodynamic tolerance to a NFB in patients with ARDS.

Table 1 (abstract 000729). See text for description

	All patients (N=35)	NFB with HT (N=18)	NFB without HT (N=17)	p
SOFA at T0	10 (7-11)	9 (6-11)	11 (8-12)	0.2
MAP at T0 (mmHg)	71 (69-72)	72 (71-73)	69 (68-72)	0.5
Heart rate at T0 (bpm)	100 (86-114)	98 (85-108)	94 (66-116)	0.2
Cardiac index at T0 (L/min/m ²) (N=27)	2.7 (2.1-3.0)	2.7 (2.4-3.1)	2.5 (2.0-3.0)	0.7
Norepinephrine dose at T0 (mcg/kg/min) (N=27)	0.2 (0.03-0.3)	0.2 (0.04-0.30)	0.1 (0.03-0.3)	0.5
PaO ₂ /FiO ₂ at T0	134 (86-193)	136 (115-200)	143 (87-198)	0.4
Time from diagnosis to initial NFB (hours)	74 (49-96)	72 (48-96)	74 (64-94)	0.6
Furosemide dose during NFB (mg/24 hours)	80 (70-120)	80 (80-120)	80 (60-120)	0.9
Negative fluid balance at T24 (L)	1.3 (1.1-2.0)	1.7 (1.2-2.0)	1.3 (1.1-1.9)	0.1

001254

Monitoring resuscitation in severe sepsis and septic shock patients. Results of the MORESS trial

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INTRODUCTION. During the resuscitation process of septic shock, use of dynamic parameters over static parameters to assess fluid responsiveness has been proposed. Surprisingly, scarce evidence is available showing a beneficial clinical outcome effect of dynamic parameters over static parameters to guide fluid resuscitation.

OBJECTIVES. To evaluate the impact on mortality of a fluid resuscitation protocol based on the use of dynamic parameters compared to static parameters to assess fluid responsiveness in the resuscitation process of severe sepsis and septic shock patients.

METHODS. Prospective, randomized, multicenter study. Adult patients with severe sepsis or septic shock, with evidence of tissue hypoperfusion, and under mechanical ventilation. At ICU admission, patients were randomized to one of the two fluid resuscitation protocols: *Control Group*, where fluid resuscitation was guided by Central Venous Pressure (CVP), according to the recommendations of *Surviving Sepsis Campaign 2012*; and *Intervention Group*, where fluid resuscitation was guided by Pulse Pressure Variation (PPV), Systolic Volume Variation (SVV), or Passive Leg Raising (PLR) maneuver. The fluid resuscitation protocol was followed during the first 72h of ICU stay. Main study outcome was 28-day mortality. Baseline patient characteristics, hemodynamic data, metabolic data, fluid balance, organ failure evolution, and ICU/hospital stay were also recorded.

RESULTS. The study was interrupted after two years because of low patient inclusion rate. A total of 60 patients were included in 12 centers. 66.6% of the patients were randomized to control group (n=40), and 33.3% to intervention group (n=20). There were no differences in baseline patients characteristics between the two groups. Median age was 62 $\pm$ 15 years, 65% male, 82% in septic shock, with initial SOFA score 12 $\pm$ 4 points. Lactate at inclusion was 4.7 $\pm$ 3.6 mmol/L. Global mortality was 30%, 47% in the control group and 21% in the intervention group ($p = 0.1$). There were no differences in lactate evolution at 6 and 24 hours of admission. Fluid balance during the study period (72h) was significantly increased in the control group (5812 $\pm$ 4818 mL vs 3035 $\pm$ 3031 mL, $p 0.02$).

CONCLUSION. A fluid resuscitation protocol based on dynamic parameters to assess fluid responsiveness is associated to a decreased fluid balance in the first 72h of the resuscitation process of septic shock. The inclusion of these dynamic variables may help to limit excessive fluid administration, potentially improving the outcome in this population of patients.

HSRO - Predicting outcome, what are we missing?

000349

Factors associated with changes in frailty one year after ICU admission; a prospective cohort study

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INTRODUCTION. Frailty significantly impacts ICU patient's outcomes, recovery and healthcare utilization. However, knowledge about changes in frailty after ICU admission and associated factors is limited.

OBJECTIVES. To assess frailty before ICU admission and determine differences between frail and non-frail patients; to study changes in frailty in the year after ICU admission and its associated variables.

METHODS. Adult patients admitted for ≥ 12 hours to a university hospital ICU between July 2016 and December 2017 were included (MONITOR-IC study, clinicaltrials.gov NCT03246334). Moribund patients were excluded. Frailty was assessed using the Clinical Frailty Scale (CFS) (range 1 'Very fit' to 9 'Terminally ill') before ICU admission (T0), at hospital discharge (T1), and three (T2) and 12 months (T3) after ICU admission. Baseline differences between 'non frail' (scale 1-4) and 'frail' patients (scale 5-9) were compared. We calculated CFS mean differences between time points. We used multiple linear regression to determine factors associated with frailty change scores over one year (T0-T3).

RESULTS. A total of 1300 patients was included: mean age 61 (± 14.9 years), 65% male and 66% planned ICU admissions. Median baseline CFS score was 3 [IQR:2-4]. Frail patients (12%) were significantly more often female, lower educated, divorced or widowed, living in a healthcare facility, had more often a chronic condition and an unplanned ICU admission.

At T1 the number of frail patients increased from 12 to 43%. At T2 and T3 this decreased to 15% and 9%. After one year, 30% had a lower CFS score compared to T0 (less frail), 31% a higher score (more frail) and in 39% the score did not change. Factors significantly associated with lower scores after 12 months, were planned admission ($B=-0.306$), mechanical ventilation ($B=-0.320$) and higher baseline CFS score ($B=-0.711$). Factors significantly associated with higher scores were age ($B=0.013$), living in a healthcare facility before admission ($B=1.199$), hospital LOS ($B=0.023$), and being discharged to a nursing home ($B=0.861$). APACHE IV score, chronic diagnosis, admission type and ICU LOS were not associated with changes in frailty.

CONCLUSION. Frailty levels changed in the first year after ICU admission. After one year, one third of the patients were more frail and one third less frail. Several factors, including age, living in healthcare